

Final Examination

1. (10 points) Let $X_1, \dots, X_n$ be n independent random variables with a same Poisson distribution with rate parameter $\lambda > 0$. Define $S_n = X_1 + \dots + X_n$.
 - (a) (3 points) Find $\mathbb{E}(S_n|X_1)$.
 - (b) (3 points) Find $\mathbb{E}(X_1|S_n)$.
 - (c) (4 points) Let s be a non-negative integer and consider the conditional distribution of $(X_1, \dots, X_n)$ given $S_n = s$.
 - (c1) What is the domain of definition of this conditional distribution?
 - (c2) Determine the p.m.f. of the conditional distribution.

2. (20 points)

Let $Z_0, Z_1, Z_2, \dots$ be independent, identically distributed random variables with common p.m.f.:

k	0	1	2	3
p_k	0.1	0.3	0.2	0.4

For $n \geq 0$, define $X_n = \min(Z_0, Z_1, \dots, Z_n)$.

- (a) (4 points) Give the transition matrix P of the Markov chain (X_n) .
 - (b) (2 points) Calculate $\mathbb{P}(X_3 = 2, X_2 = 1, X_1 = 2|X_0 = 3)$
 - (c) (2 points) Calculate $\mathbb{P}(X_3 = 0, X_2 = 1, X_1 = 2|X_0 = 3)$.
 - (d) (2 points) Find the communication classes of the chain.
 - (e) (2 points) Find the recurrent and transient states of the chain.
 - (f) (2 points) What are the periods of the states?
 - (g) (3 points) Does the chain have a stationary (invariant) distribution?
 - (h) (3 points) Find the limiting probabilities of the chain.
3. (10 points) Let X_1, X_2 and X_3 be independent exponential random variables with respective rates $\lambda_j, j = 1, \dots, 3$.
 - (a) (5 points) Show that

$$\mathbb{P}(X_1 < X_2 < X_3) = \frac{\lambda_1 \lambda_2}{(\lambda_2 + \lambda_3)(\lambda_1 + \lambda_2 + \lambda_3)}.$$

- (b) (5 points) What is the value of this probability when all the λ_j 's are equal? Can you find this value by a different method?
4. (15 points) Let $\{N_t\}_{t \geq 0}$ be a Poisson process with rate 3.
 - (a) (2 points) Calculate $P(N_4 = 3)$.
 - (b) (2 points) Calculate $P(N_4 = 3, N_6 = 6, N_{10} = 12)$.
 - (c) (2 points) Calculate $P(N_6 = 6|N_4 = 3)$.
 - (d) (2 points) Calculate $P(N_6 = 6|N_{10} = 12)$.
 - (e) (2 points) Determine $E(N_{10}|N_6)$ and $E(N_{10}|N_4, N_6)$.
 - (f) (5 points) Show that conditional on $N_{10} = n$, N_6 has the binomial distribution $\mathcal{B}(n, 3/5)$. Determine $E(N_6|N_{10})$.

5. (10 points) An insurance company has two insurance portfolios, namely A and B . Claims in portfolio A occur in accordance with a Poisson process with mean 3 per year. Claims in portfolio B occur in accordance with a Poisson process with mean 5 per year. The two processes are independent.
- (5 points) Calculate the probability that the 3rd claim in portfolio A occurs before the 3rd claim in portfolio B in a certain year.
 - (5 points) Given that exactly eight claims had occurred in the first year, find the expected arrival time of the first claim.
6. (15 points) Consider a special two-station open Jackson network. Each station has a single server and an infinite capacity waiting area. External arrivals to station 1 follow a Poisson process with rate α , and service times at station i are i.i.d, following exponential distribution with rate μ_i , $i = 1, 2$. After being served at station 1, each customer goes to station 2, and after being served at station 2, the customer goes back to station 1 with probability 20% and leaves the system with probability 80%.
- Assume that
- $$(1.25)\alpha < \mu_1 \text{ and } (1.25)\alpha < \mu_2.$$
- Let $X_i(t)$ be the number of customers at station i at time t , $i = 1, 2$. Then $X = \{(X_1(t), X_2(t)), t \geq 0\}$ is a continuous time Markov chain.
- (5 points) Prove that X is irreducible.
 - (10 points) Prove that X has a stationary distribution by the following three steps:
 - write the flow balance equation for *each state* according to the four cases: $(0, 0)$, $(n_1, 0)$, $(0, n_2)$ and (n_1, n_2) where $n_1 > 0$ and $n_2 > 0$;
 - guess a form of the stationary distribution, and
 - verify that the guessed stationary distribution does satisfy the flow balance equation for each state.
7. (20 points) Let $B(t), t \geq 0$ be a standard Brownian motion.
- (5 points) Determine $P(B(2.1) > 1.7 | B(1.6) = 0.5)$
 - (5 points) Determine $P(B(1.6) \leq 0.5 | B(2.1) = 1.7)$
 - (5 points) Show that for any $0 < s < t$, $B(t)$ and $B(s) - \frac{s}{t}B(t)$ are independent.
 - (5 points) Let $M(t) = \max_{0 \leq s \leq t} B(s)$. Determine $\mathbb{E}[M(1)]$.

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